

Owusu Kantankah

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EDUCATION

University of Wisconsin-Madison

Madison, WI

Bachelor of Science in Computer Science and Data Science

EXPECTED May 2028

Cumulative GPA: 3.43

Relevant Coursework: High Performance Computing (Graduate), Computer Graphics, Algorithms

PROFESSIONAL EXPERIENCE

LASER Research Program, UW-Madison

Madison, WI

Undergraduate Machine Learning Researcher

Summer 2026

- Earned a **return offer** to continue as an **ML researcher** through the school year and a **Teaching Assistant** appointment for **Computer Graphics** in the spring, based on research performance this summer.
- Engineered a multi-agent actor-critic RL pipeline (PyTorch/CUDA) with generalized advantage estimation, simulating up to **250** drone agents to compare centralized, fully independent, and locally-communicating MARL architectures on training efficiency and convergence in a shared flocking task
- Profiled and optimized the training loop's GPU utilization, replacing per-agent-group Python-looped linear layers with a single batched matrix multiply and applying torch.compile/CUDA graph capture to the rollout path for a **2.3x** per-episode speedup (**0.884s** → **0.384s**)

Space Science and Engineering Center

Madison, WI

Undergraduate Data Engineering Researcher

September 2025 - May 2026

- Built Python data pipeline (NumPy, pandas) processing **10GB+** of NetCDF spacecraft telemetry from JAXA's Akatsuki orbiter, cutting analysis runtime by **35%** and enabling faster iteration.
- Identified a **~195K** polar vortex and **20K** equator-to-pole thermal gradient via zonal-mean analysis on 1440x720 NetCDF grids, contributing to ongoing research on Venus cloud-top absorbers.
- **Presented at UW-Madison's Undergraduate Research Symposium.**

StudioNil

Remote

Founder & Lead Software Developer

March 2023 - August 2025

- Founded and shipped a multiplayer game that reached **10,000** MAU in 6 weeks, architected client-server infrastructure handling **100** concurrent players per instance at **0.02%** crash rate.
- Engineered a custom 3D spatial hitbox system using raycasting and geometric detection, replacing Roblox's native .Touched API and improving collision accuracy by **1.8x** in real-time combat.

PROJECTS

CPU Raytracer [\[Link\]](#) - C++, GCC, CMake, Git

June 2026

- Built a CPU-based ray tracer in C++ implementing ray-sphere intersection, diffuse/metal/dielectric materials, reflections, refractions, antialiasing, and depth-of-field camera effects.

3D Drone Simulation [\[Link\]](#) - Three.js, GLSL, Javascript, HTML, CSS, Git

May 2026

- Built a 3D drone swarm simulation in Three.js with custom GLSL shaders, implementing Reynolds' boids algorithm (separation, cohesion, alignment) to drive emergent flocking across **1000** agents.
- Collapsed **1000** per-drone draw calls into a single InstancedMesh dispatch with per-instance transform matrices, improving FPS from **80** to **133**, and halving GPU memory to **9 MB**.

Centralized Morgridge Event Calendar [\[Link\]](#) - Python, Flask, SQLite, AWS, Git

February 2026

- Built RESTful Flask API with SQLite database supporting event retrieval, insertion, and channel management, deployed on AWS EC2 as the data layer between automated scrapers and a React frontend; contributed as backend engineer on a **9-person** team.

SKILLS

- **Languages:** Java, Python, C++, C, Lua, GLSL, SQL, TypeScript/Javascript, HTML/CSS
- **Frameworks & Libraries:** PyTorch, NumPy, pandas, Matplotlib, Three.js, React, Flask, Tailwind
- **Developer Tools:** Git, GitHub, Linux, AWS (EC2), SQLite, CMake, Roblox Studio, Visual Studio Code